

Lizhe Zhang

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Research Interests: Agent Safety & Security • LLM Evaluation • Trustworthy AI

EDUCATION

University of California, Santa Cruz, Santa Cruz, CA

Ph.D. in Computer Science and Engineering

Sep. 2026 – Present

Advisor: Prof. Chenguang Wang

University of California, San Diego, La Jolla, CA

M.S. in Computer Science

Sep. 2024 – Mar. 2026

GPA: 4.00/4.00

Emory University & Georgia Institute of Technology, Atlanta, GA

B.S. in Physics, Minor in Applied Mathematics (Emory)

B.S. in Computer Engineering (Georgia Tech)

Aug. 2019 – Dec. 2023

GPA: 3.90/4.00

GPA: 3.96/4.00

RESEARCH EXPERIENCE

University of California, Santa Cruz

Graduate Student Researcher

Sep. 2026 – Present

Advisor: Prof. Chenguang Wang

- Investigating the safety and security of LLM-based agents, with a focus on agentic attacks, tool use, and robust agent behavior.

University of California, San Diego

Graduate Researcher

Jun. 2024 – Mar. 2026

Advisor: Prof. Jingbo Shang

- Developed an automated code-rewriting framework for diagnosing harmful memorization in LLM code generation, and introduced the **Harmful Memorization Index (HMI)** to distinguish brittle reuse of familiar solution patterns from benign code reuse.
- Evaluated Qwen and Llama model families on MBPP+ and BigCodeBench, finding that harmful memorization does not monotonically increase with model scale and that PPO yields a more favorable accuracy–HMI trade-off than SFT.

Rice University

Graduate Researcher

Sep. 2024 – May 2025

Advisor: Prof. Xia Hu

- Developed an uncertainty-based routing framework that dynamically delegates queries from on-device small language models to larger models, balancing prediction accuracy, inference latency, and cost.
- Benchmarked 8 small LMs, 2 large LMs, and 8 uncertainty-estimation methods across reasoning and knowledge tasks, and developed a data-agnostic proxy routing strategy that generalizes to unseen datasets.

Georgia Institute of Technology

Undergraduate Researcher

May 2023 – Feb. 2024

Advisor: Prof. Thomas Plötz

- Contributed to **IMUGPT 2.0** and wearable emotion-recognition research by implementing and running virtual-IMU generation pipelines, conducting large-scale experiments across multiple motion datasets, and preparing synthetic sensor data for downstream training and evaluation.

PUBLICATIONS

Evaluating Harmful Memorization in LLM Code Generation

*Lizhe Zhang**, Wentao Chen*, Li Zhong, Letian Peng, Zilong Wang, Jingbo Shang

EMNLP 2026 Findings

[\[paper\]](#)

Confident or Seek Stronger: Exploring Uncertainty-Based On-device LLM Routing From Benchmarking to Generalization

*Yu-Neng Chuang**, Leisheng Yu*, Guanchu Wang, *Lizhe Zhang*, Zirui Liu, Xuanning Cai, Yang Sui, Vladimir Braverman, Xia Hu

NeurIPS 2025 Workshop

[\[paper\]](#)

IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition

Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, *Lizhe Zhang*, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz

IMWUT 2024

[\[paper\]](#)

Emotion Recognition on the Go: Utilizing Wearable IMUs for Personalized Emotion Recognition

Zikang Leng, Myeongul Jung, Sungjin Hwang, Seungwoo Oh, *Lizhe Zhang*, Thomas Plötz, Kwanguk Kim

UbiComp/ISWC 2024

[\[paper\]](#)

WORK EXPERIENCE

Machine Learning Intern

Horizon Robotics

May 2023 – Aug. 2023

Beijing, China

- Evaluated state-of-the-art 3D semantic segmentation models on SemanticKITTI, benchmarking their performance for autonomous-driving perception.
- Identified failure modes of standard mIoU when evaluating small spatial deviations in 3D predictions and collaborated with the perception team to prototype a range-aware evaluation framework for more informative model comparison.

Software Engineering Intern

Inspur Group

Jun. 2021 – Aug. 2021

Jinan, China

- Replaced a legacy Haar-cascade license plate detector with a fine-tuned YOLOv5 model, improving detection accuracy by over 40% and increasing robustness under diverse and low-light conditions.

SELECTED PROJECT

Find Your Spot

Georgia Institute of Technology, Senior Design Project

Aug. 2023 – Dec. 2023

[\[poster\]](#)

- Built an end-to-end indoor parking navigation system using BLE beacons and RSSI-based localization to guide drivers to available parking spaces.
- Developed the backend for processing IoT sensor and beacon data, integrating localization and routing algorithms for real-time parking navigation.

PATENTS & HONORS

Language-based Cross-modality Transfer System for Generating Virtual Inertial Measurement Unit Data to Enhance Human Activity Recognition

U.S. Patent Application No. 19/566,099

Filed Mar. 2026

Sigma Pi Sigma National Physics Honor Society

Emory University, 2023

Faculty Honors

Georgia Institute of Technology, 2022–2023

ACADEMIC SERVICE

Organizer, Agents in the Wild: Safety, Security, and Beyond

Proposed for ICLR 2027

Reviewer, Conference on Language Modeling (COLM)

2026

TEACHING EXPERIENCE

Teaching Assistant, CS 170: Introduction to Computer Science, Emory University

Fall 2021, Spring 2022

SKILLS

Programming: Python, C/C++, Java, Swift

Machine Learning: PyTorch, Hugging Face Transformers, TensorFlow, scikit-learn, OpenCV

Tools: Git, Docker, vLLM, LaTeX